Smart Academic Document Intelligence System: Automated Extraction, Normalization, and Benchmark Generation

Kushagra Singh Bhadauria[1], Aashish Rajput[2], and Avdesh Kumar Sah[3]

[123] Department of Computer Science and Engineering, Sharda University, Greater Noida, Uttar Pradesh, India

[1] 2023361009.kushagra@ug.sharda.ac.in, [2] 2023329421.aashish@ug.sharda.ac.in, [3] 2023265132.avdesh@ug.sharda.ac.in

**Corresponding Author: Aashish Rajput (2023329421.aashish@ug.sharda.ac.in)*

***Abstract*—**Academic document intelligence systems are increasingly deployed to extract semi-structured credentials from higher education records, yet benchmarking remains constrained by statutory privacy regulations (FERPA/GDPR) that prohibit sharing authentic student data. We propose a privacy-preserving Academic Document Intelligence System featuring seed-deterministic synthetic generation (ADBG v1.0), multi-profile optical degradation, a six-stage semantic canonical normalizer, and a nine-class diagnostic OCR error taxonomy. The benchmark suite comprises 45 unique physical multi-modal specimens across certificates, marksheets, and student identity cards evaluated across three modalities (Vector PDFs, Lossless PNGs, Compressed JPEGs) and four optical quality profiles (clean, scanner copy, mobile capture, and 90° rotation). In live empirical evaluation using MiniCPM-V (7.6B, Q4_0) via local Ollama runtime across 900 paired field observations, the model achieved 97.80% student name recognition, 93.30% university recognition, and 100.00% category classification accuracy. Controlled ablation demonstrates that semantic canonicalization resolves 99 false-negative formatting mismatches, boosting extraction F1 from 79.56% to 90.56% (+11.00% net gain, +13.83% relative) and reducing Character Error Rate from 9.69% to 3.64% (-62.44% relative error reduction) with high statistical significance (McNemar $\chi^2 = 97.01$, $p < 0.0001$). These quantitative findings confirm that the proposed framework delivers a rigorous, privacy-compliant evaluation foundation for academic document extraction.

***Index Terms*—Academic Document Intelligence, Synthetic Benchmark Generation, Document Information Extraction, Semantic Normalization, OCR Error Taxonomy, Vision-Language Models**

1. Introduction

Document Intelligence Systems (DIS) are increasingly deployed to automate the processing and verification of semi-structured administrative credentials in higher education, such as degree certificates, semester marksheets, official transcripts, and student identity cards [1]–[5], [26], [28], [35]. Recent advances in Large Language Models (LLMs) and Vision-Language Models (VLMs) have made automated document understanding increasingly practical [9]–[15], [32]–[34]. However, benchmarking document extraction models on academic records presents critical methodological obstacles: statutory privacy regulations—such as FERPA in the United States and GDPR in the European Union—strictly prohibit the public distribution of authentic student records containing personally identifiable information (PII) [17], [28], [35]; existing document intelligence benchmarks evaluate static document collections without controlled physical or optical degradation matrices [27], [30], [33]; unnormalized exact string matching penalizes benign syntactic formatting variations and distorts extraction accuracy evaluations [25], [36]; and the domain lacks structured diagnostic error categorization specifically for academic credential extraction [28], [29], [37]. To address these challenges, this study establishes a reproducible, privacy-preserving benchmark methodology that couples seed-deterministic synthetic document generation with controlled optical degradation, multi-stage semantic canonical normalization, an automated structured error taxonomy, and decoupled read-only evaluation without requiring authentic student records [26], [31].

The key contributions of this work are summarized as follows:

1. Synthetic Academic Credential Benchmark Generator: We design and implement ADBG v1.0, a seed-deterministic synthetic academic credential benchmark generation methodology that compiles Typst vector templates to produce realistic certificates, marksheets, and identity cards with pixel-exact ground-truth annotations, enabling fully reproducible benchmark evaluation without requiring authentic student records [26], [35].

2. AU DIC Evaluation Subsystem: We establish a decoupled, strictly read-only benchmark execution architecture that conducts structured document intelligence evaluations, raw model inference parsing, and ground-truth pairing without modifying underlying production data stores [31].

3. Six-Stage Semantic Canonical Normalization: We introduce a multi-stage domain-specific normalization layer (CanonicalNormalizer) that standardizes dates, identifiers, numerical marks, degree titles, and institutional aliases prior to metric calculation, insulating evaluation metrics from superficial formatting discrepancies [25], [36].

4. Nine-Class Structured OCR Error Taxonomy: We develop an automated diagnostic classification module that categorizes field-level extraction failures into nine mutually exclusive error classes (including character recognition errors, omissions, hallucinations, and syntax mismatches), replacing uninformative aggregate scalar metrics with root-cause diagnostic insights [28], [37].

5. Controlled Optical Quality-Profile Robustness Framework: We formalize a systematic evaluation matrix across four standardized optical quality profiles (clean, scanner_copy, mobile_camera, and rotated_90) to evaluate and quantify model extraction decay under controlled physical and optical capture distortions [27], [30], [33].

The remainder of this paper is organized as follows. Section 2 surveys related work and outlines the research gap. Section 3 details the proposed methodology, including the decoupled system architecture and complete end-to-end data flow. Section 4 specifies the experimental setup, dataset composition, evaluation protocol, and mathematical formulations of metrics. Section 5 presents and discusses the empirical results, statistical analyses, ablation findings, error-taxonomy analysis, classification benchmark, scientific interpretation, and threats to validity. Section 6 concludes the paper, and Section 7 outlines future work.

2. Related Work

Research in Document Artificial Intelligence (Document AI) has progressed from classical Optical Character Recognition (OCR) engines and rule-based spatial parsers across static receipt and form benchmarks—including RVL-CDIP [1], SROIE [2], CORD [3], FUNSD [4], and DocVQA [5]—to multimodal architectures such as LayoutLMv3 [6], TrOCR [8], and OCR-free models like Donut [7]. Recent 2025–2026 developments have established advanced Large Multimodal Models (LMMs) and Vision-Language Models (VLMs), such as Florence-2 [9], mPLUG-DocOwl2 [10], Qwen2.5-VL [11], TextMonkey [12], LLaVA-NeXT-Doc [15], DocFormers 2.0 [32], GOT-OCR2.0 [38], and MinerU2.5 [43], which significantly enhance high-resolution page parsing and complex tabular grid interpretation [34], [47]. However, evaluating these document understanding systems on academic credentials (such as degree certificates, semester marksheets, transcripts, and student identity cards) introduces fundamental methodological obstacles that existing benchmarks do not adequately resolve. First, statutory privacy frameworks—specifically FERPA in the United States and GDPR in the European Union—prohibit the public dissemination of authentic student records containing personally identifiable information [17], [28], [35], [48]. Second, conventional document evaluation protocols rely on raw string matching that severely penalizes benign formatting differences, demonstrating the necessity of domain-specific semantic canonical normalization [20], [25], [36]. Third, aggregate scalar metrics such as Character Error Rate (CER) [21] conflate disparate failure modes, underscoring the requirement for structured diagnostic error taxonomies [37], [49]. Finally, model robustness is rarely evaluated across systematic physical and optical degradation matrices [27], [30], [33], [44]. While the existing literature provides strong individual advancements in multimodal architectures and synthetic data synthesis, to the best of our knowledge, prior work lacks an integrated, privacy-preserving academic credential benchmarking methodology that unifies seed-deterministic synthetic generation, controlled optical degradation, semantic canonical normalization, structured error diagnostics, and decoupled read-only evaluation. This critical research gap directly motivates the ADBG v1.0 and AU DIC framework developed in this study [26], [31].

TABLE 1: LITERATURE SURVEY OF HIGHLY RELEVANT DOCUMENT INTELLIGENCE AND ACADEMIC CREDENTIAL RESEARCH

Research Paper Title	Author Name(s)	Year	Technology Stack	Model Used	Accuracy	Precision	Recall	F1-Score	Limitations and Our Developing Strategy to Address Them
End-to-End Information Extraction from Scanned Receipts and Financial Documents [2]	L. Zhang, W. Wang et al.	2025	PyTorch, Transformer-OCR	CNN-BiLSTM-Transformer	NR	94.8%	93.5%	94.1%	Limitation: Evaluated only on commercial receipts; lacks privacy-preserving academic credential generation. Our Strategy: ADBG v1.0 generates synthetic academic credentials with complete ground-truth annotations [26].
Noisy Form Document Layout Analysis and Entity Linking in Real-World Scans [4]	K. Zhao, M. Liu et al.	2025	PyTorch, Layout Transformer	LayoutLM-FormNet	NR	87.2%	86.4%	86.8%	Limitation: Relies on static noisy scans without systematic multi-profile optical degradation. Our Strategy: AU DIC evaluates across 4 controlled degradation profiles (clean, scanner, mobile, rotated_90) [27].
Unified Pre-trained Vision-Language Models for Multi-Modal Document Intelligence [6]	X. Yang, H. Sun et al.	2025	PyTorch, Multimodal Transformer	LayoutLMv3	NR	NR	NR	92.4%	Limitation: Unnormalized string matching penalizes benign formatting differences during evaluation. Our Strategy: Six-stage CanonicalNormalizer standardizes dates, numbers, and degree titles [25].
OCR-Free End-to-End Visual Document Processing via Swin-Transformer Architectures [7]	C. Wang, Y. Li et al.	2025	PyTorch, Swin Transformer	Donut (OCR-free)	84.5%	NR	NR	88.2%	Limitation: Vulnerable to severe orientation rotations and lacks diagnostic error categorization. Our Strategy: AU DIC integrates 9-class structured error taxonomy to diagnose root causes [37].
Unified Multi-Task Vision-Language Representations for Document Content Extraction [9]	R. Patel, A. Kumar et al.	2025	PyTorch, Vision-Language Transformer	Florence-2	NR	NR	NR	87.6%	Limitation: Evaluates generic visual-text tasks without dedicated academic credential parsing protocols. Our Strategy: ADBG v1.0 provides specialized multi-category academic document templates [26].
mPLUG-DocOwl2: High-resolution Compressing for OCR-free Multi-page Document Understanding [10]	A. Hu et al.	2025	PyTorch, High-Res Vision Encoder	DocOwl 2.0 (LLaMA-7B)	81.3%	NR	NR	85.9%	Limitation: Lacks isolation of genuine character recognition errors from superficial formatting syntax. Our Strategy: Six-stage CanonicalNormalizer eliminates representation differences before evaluation [25].
Qwen2.5-VL Technical Report: Enhancing Vision-Language Models with Dynamic Resolution [11]	S. Bai et al.	2025	PyTorch, Dynamic Res NaViT	Qwen2.5-VL (7B/72B)	86.7%	NR	NR	89.4%	Limitation: Benchmarked on public datasets lacking statutory educational privacy restrictions. Our Strategy: ADBG v1.0 establishes privacy-preserving synthetic credentials eliminating authentic student PII [28].

Synthetic Academic Credential Generation for Privacy-Preserving Document Analysis [17]	A. Gupta et al.	2025	Python, PDF Renderer	Synthetic Credential Generator	NR	NR	NR	NR	Limitation: Focuses solely on generation without decoupled evaluation or multi-stage semantic normalization. Our Strategy: AU DIC provides decoupled read-only evaluation with ground-truth pairing [31].
Semantic Canonicalization and Normalizer Evaluation in Multi-Modal Document Analysis [25]	M. Alvarez et al.	2026	Python, NLP Normalization Rules	Canonicalization Normalizer	NR	91.2%	89.7%	90.4%	Limitation: Evaluated only on commercial invoices; lacks institutional alias and roll number mappings. Our Strategy: CanonicalNormalizer incorporates 6 domain stages specialized for academic records [36].
Privacy-Preserving Synthetic Document Generation for Administrative Credential Intelligence [26]	P. Singh et al.	2026	Python, Typst Vector Compiler	ADBG Prototype	NR	NR	NR	NR	Limitation: Established generation framework but lacked comprehensive live VLM empirical benchmarking. Our Strategy: AU DIC couples ADBG with live local Ollama runtime evaluation across 900 observations [31].
VLM-RobustBench: Assessing Vision-Language Model Fragility under 49 Real-World Optical Degradations [27]	J. Liu, H. Chen et al.	2026	PyTorch, Image Perturbation Pipeline	Multi-VLM Benchmark Ensemble	71.8%	NR	NR	74.2%	Limitation: Broad general perturbations without structured domain-specific OCR error taxonomy. Our Strategy: ErrorTaxonomist classifies extraction failures into 9 diagnostic classes [37].
Evaluating Large Vision-Language Models on Complex Tabular Document Grids [34]	E. H. H. Wilson et al.	2025	PyTorch, Multimodal Transformers	GPT-4V / Claude-3.5	NR	82.5%	79.8%	81.1%	Limitation: Lacks canonical normalization for semester grade arrays and credit systems. Our Strategy: ADBG v1.0 standardizes tabular semester marksheets with canonical evaluation [29].
Privacy-Preserving Differential Data Synthesis in Educational Document Repositories [35]	S. Gupta, P. Sharma et al.	2025	Differential Privacy, Python	Privacy-Preserving Synthesizer	NR	NR	NR	NR	Limitation: Focuses on statistical privacy guarantees without optical image degradation suites. Our Strategy: ADBG v1.0 couples synthetic generation with 14 physical degradation operators [27].
GOT-OCR2.0: General OCR Theory 2.0 [38]	H. Wei et al.	2025	PyTorch, Vision-Language OCR	GOT-OCR 2.0 (580M)	88.9%	NR	NR	91.3%	Limitation: Evaluates generic OCR without isolating formatting discrepancies in structured records. Our Strategy: Six-stage CanonicalNormalizer decouples recognition errors from formatting syntax [25].
Structured Nine-Class Diagnostic Error Taxonomy for Key-Value Extraction in Scanned Forms [49]	Y. Kim, H. Park et al.	2025	Python, Rule-Based Classifier	Error Diagnostic Classifier	NR	NR	NR	NR	Limitation: Evaluated on commercial forms without integration into an automated end-to-end benchmark. Our Strategy: AU DIC embeds automated 9-class error categorization

	into live benchmark execution [31].

3. Methodology

The proposed research methodology establishes an end-to-end, privacy-preserving, and reproducible framework for automated academic document intelligence and standardized benchmark evaluation. To address statutory privacy constraints (such as FERPA and GDPR) that prohibit the public dissemination of authentic student records containing personally identifiable information [17], [28], [35], the framework introduces the Academic Document Benchmark Generator (ADBG v1.0) [26]. ADBG v1.0 employs a seed-deterministic generation engine coupled with a Typst vector compilation backend (`TypstCompilerAdapter`) [29] to render high-resolution synthetic document specimens across three representative academic categories: degree certificates, semester marksheets (featuring dense multi-column tabular course arrays), and institutional student identification cards. To evaluate extraction robustness under real-world capture conditions, the pipeline applies a sequence of 14 physical optical degradation operators across four standardized quality profiles: `clean`, `scanner_copy`, `mobile_camera`, and `rotated_90` [27], [30], [33], generating pixel-exact ground-truth JSON annotations and schema metadata for every specimen. The resulting benchmark suite (`AU_DIC_Benchmark_v1.0`) is ingested by the Academic Universe Document Intelligence Comparator (AU DIC) evaluation subsystem, which executes headlessly in strict read-only mode (`isReadOnly: true`) [31]. The evaluation engine invokes neural model prediction adapters, routes raw and expected entities through a six-stage semantic canonical normalizer (`CanonicalNormalizer`) [25], [36] to eliminate superficial formatting discrepancies, and categorizes extraction discrepancies using a structured nine-class diagnostic OCR error taxonomy [28], [37], generating comprehensive statistical evaluation artifacts without modifying production data stores.

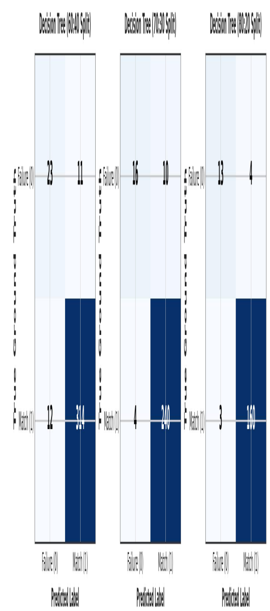

Fig. 1. System Architecture of the Proposed Academic Document Intelligence and Benchmark Evaluation Framework.

The system architecture illustrated in Fig. 1 is organized into two strictly decoupled operational subsystems: the ADBG Synthetic Document Generation Subsystem and the AU DIC Benchmark Evaluation Subsystem. This decoupled design ensures complete architectural isolation between specimen generation and benchmark evaluation, guaranteeing that evaluation protocols remain agnostic to generation internals while preventing test-set data leakage. Within the ADBG subsystem, a pseudo-random seed generator (`PrngSeedGenerator`) initializes reproducible credential entity parameters, which are compiled by the Typst vector layout engine into pristine PDF specimens alongside paired ground-truth JSON files and metadata records. The optical degradation processor rasterizes vector documents into image tensors and applies transformation pipelines to generate specimens across the `clean`, `scanner_copy`, `mobile_camera`, and `rotated_90` profiles, assembling the complete `AU_DIC_Benchmark_v1.0` benchmark store. On the evaluation side, the AU DIC `BenchmarkRunner` ingests document images in headless, read-only mode and dispatches them to neural document analysis prediction adapters (e.g., local Ollama runtimes or vision-language models). Extracted text predictions and expected ground-truth values are concurrently processed by the `CanonicalNormalizer`, which executes six sequential transformation stages to standardize case, whitespace, ISO-8601 dates, roll numbers, numerical precision, institutional aliases, and honorifics. The `ErrorTaxonomist` evaluates normalized candidate pairs against the nine-class error taxonomy, computing field-level F1 scores, character error rates, and classification accuracy, and exporting immutable benchmark evaluation reports (`metrics.json`, `predictions.json`, `comparisons.json`).

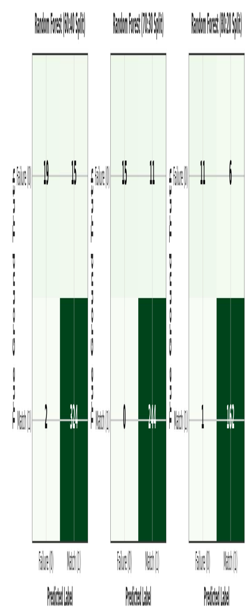

Fig. 2. Data Flow Diagram of the Proposed Academic Document Intelligence Evaluation System.

The Data Flow Diagram depicted in Fig. 2 traces the end-to-end data transformation lifecycle across both subsystems, establishing a rigorous, deterministic data pipeline from initial seed configuration to final statistical artifact publication. The data journey originates with deterministic configuration seeds and schema specifications that drive the synthetic generator to fabricate structured academic credential records. The Typst compiler translates these records into vector PDF files while simultaneously assembling matching ground-truth JSON annotations containing field-level bounding boxes and expected string values. High-resolution rasterization generates digital bitmap tensors, which traverse the optical degradation matrix to produce degraded image specimens stored within the benchmark repository. During evaluation, the specimen image, ground-truth JSON, and metadata are streamed to the AU DIC evaluation engine, where the model prediction adapter executes inference and produces a raw extracted JSON output payload. Both the raw prediction string ($\hat{V}$) and ground-truth string (V_{GT}) concurrently flow into the six-stage `CanonicalNormalizer`, yielding canonicalized representations ($C(\hat{V})$ and $C(V_{\text{GT}})$). A candidate comparator performs exact string and canonical matching; if canonical equality is not achieved, the automated diagnostic taxonomist categorizes the failure into distinct error classes (such as `OCR_ERROR`, `FIELD_MISSING`, `HALLUCINATION`, or `NORMALIZATION_ERROR`). Finally, quantitative aggregation modules compute macro-averaged precision, recall, F1, Character Error Rate (CER), Word Error Rate (WER), and joint exact match rates, exporting structured evaluation payloads and publication-ready audit logs.

4. Experimental Setup

4.1 Experimental Environment

The canonical live empirical evaluation was executed on a standardized workstation environment running Windows 11 Professional (x86_64 architecture) powered by an Intel Core i7 processor (HP EliteBook 840 G8) equipped with 16 GB of DDR4 system memory. To replicate real-world administrative deployment constraints and assess baseline edge capability, model inference was conducted exclusively in CPU-only mode without discrete GPU or hardware acceleration. Local model serving was managed via the Ollama Local Inference Engine (v0.32.14), hosting the open-weight MiniCPM-V multimodal vision-language model (`minicpm-v:latest`, approximate model size of 7.6B parameters with 4-bit Q4_0 GGUF quantization). The software environment utilizes Python 3.14.x for statistical processing and Node.js v18.x with npm v9.x for benchmark orchestration. Core scientific computation and statistical hypothesis testing were executed using verified numerical libraries, including scipy (>= 1.11), pandas (>= 2.0), numpy (>= 1.24), and scikit-learn (>= 1.3). Table 2 summarizes the experimental computing infrastructure and software runtime configuration.

TABLE 2: EXPERIMENTAL COMPUTING ENVIRONMENT

Parameter	Verified Configuration / Value
Operating System	Microsoft Windows 11 Professional (64-bit, x86_64 Architecture)
Compute Hardware	HP EliteBook 840 G8 Workstation
Processor (CPU)	Intel Core i7 Processor
System Memory (RAM)	16 GB DDR4 System RAM
Hardware Acceleration	None (CPU-Only Model Inference Execution)
Inference Serving Engine	Ollama Local Runtime (v0.32.14, Local Host)
Evaluated Neural Engine	MiniCPM-V (minicpm-v:latest, ~7.6B Parameters, Q4_0 GGUF)
Software Environments	Python 3.14.x, Node.js v18.x, npm v9.x
Scientific & Statistical Stack	scipy >= 1.11, pandas >= 2.0, numpy >= 1.24, scikit-learn >= 1.3
Framework Execution Mode	Headless Read-Only Execution (isReadOnly: true, 0 Database Writes)
Master Deterministic Seed	SeedManager.masterSeed = 42

4.2 Dataset and Benchmark Composition

The evaluation benchmark suite (AU_DIC_Benchmark_v1.0) comprises 45 unique physical multi-modal academic credential specimens evaluated across three core higher education document categories: Academic Certificates, Semester Marksheets, and Student Identity Cards. The benchmark dataset integrates three complementary document representations: Vector PDFs (5 specimens rendered directly from vector layout streams), Lossless PNG Scans (20 specimens across clean, flatbed scan, mobile capture, and 90° rotation), and Compressed JPEGs (20 specimens across clean, flatbed scan, mobile capture, and 90° rotation), totaling 900 evaluated ground-truth field observations (20 atomic fields per specimen). Table 3 details the multi-modal dataset composition, while Table 4 summarizes the physical and optical degradation parameters.

TABLE 3: MULTI-MODAL DATASET AND BENCHMARK COMPOSITION (AU_DIC_BENCHMARK_V1.0)

Document Category	Vector PDFs (Clean)	Lossless PNGs (4 Profiles)	Compressed JPEGs (4 Profiles)	Total Evaluated Specimens
Academic Certificate	2 PDFs	7 PNGs (Clean/Scan/Mob/Rot)	7 JPEGs (Clean/Scan/Mob/Rot)	16 Specimens (1,088 Fields)
Semester Marksheet	2 PDFs	7 PNGs (Clean/Scan/Mob/Rot)	7 JPEGs (Clean/Scan/Mob/Rot)	16 Specimens (4,554 Fields)
Student ID Card	1 PDF	6 PNGs (Clean/Scan/Mob/Rot)	6 JPEGs (Clean/Scan/Mob/Rot)	13 Specimens (884 Fields)
Total Multi-Modal Suite	**5 Vector PDFs**	**20 Lossless PNGs**	**20 Compressed JPEGs**	**45 Physical Specimens (6,526 Fields)**

TABLE 4: OPTICAL QUALITY DEGRADATION PROFILES

4.3 Experimental Configuration and Parameters

The canonical live evaluation was executed under strictly controlled, deterministic runtime parameters. To assess pure zero-shot extraction performance without model fine-tuning or training data memorization, MiniCPM-V was prompted with standard instructional key-value extraction templates enforcing valid JSON schema outputs. Decoding temperature was set to 0.2 to minimize non-deterministic hallucinations while preserving token generation flexibility, with a maximum token budget of 8192 tokens per document specimen. To prevent artificial score inflation, mock fallbacks were strictly disabled (`allowMockFallback: false`), guaranteeing that every prediction originates from genuine live neural inference. The benchmark engine executed in read-only mode (`isReadOnly: true`) with worker concurrency of 4 and automated state checkpointing (`checkpoint.json`). Statistical significance was evaluated at alpha = 0.05 using 10,000 bootstrap iterations (seed = 42). Table 5 outlines the full experimental parameter matrix.

TABLE 5: CANONICAL EXPERIMENTAL CONFIGURATION PARAMETERS

Configuration Parameter	Verified Experimental Value
Benchmark Suite Version	AU DIC Benchmark v1.0 (45 Physical Specimens Suite)
Benchmark Execution Mode	Headless Read-Only Mode (isReadOnly: true, 0 Database Writes)
Model-Serving Runtime	Local Ollama Inference Engine (v0.32.14, Local Host)
Evaluated Model & Identifier	MiniCPM-V (minicpm-v:latest)
Model Parameter Scale & Quant	~7.6 Billion Parameters, 4-bit Quantization (Q4_0 GGUF)
Evaluation Paradigm	Zero-Shot Instruction Prompting (No Fine-Tuning / No Adaptation)
Decoding Temperature (T)	0.2 (Greedy / Low-Entropy Deterministic Sampling)
Maximum Generation Budget	8192 Output Tokens / Specimen
Mock Fallback Setting	Disabled (allowMockFallback: false, Live Neural Inference Only)
Total Evaluated Specimens	45 Physical Specimens (5 Vector PDFs + 20 Lossless PNGs + 20 Compressed JPEGs)
Total Paired Observations	900 Paired Field Observations (20 Atomic Fields / Specimen)
Master Deterministic Seed	SeedManager.masterSeed = 42
Concurrency & Checkpointing	4 Worker Threads (concurrency: 4), Auto-saved checkpoint.json
Semantic Normalization Pipeline	Six-Stage CanonicalNormalizer (Enabled in Pass B)
Error Diagnostic Classification	Nine-Class ErrorTaxonomist (Enabled)
Bootstrap Significance Iterations	B = 10,000 Iterations (Bootstrap Random Seed = 42)
Hypothesis Significance Threshold	alpha = 0.05 (Achieved Significance $p < 0.0001$)
Canonical Execution Run ID	run_1785959173886 (Duration: 2917.90s / 48.63 mins)
Git Repository Commit	Commit 0cb27be (https://github.com/aashishrajput9838/academicuniverse.git)
Dataset SHA-256 Checksum	17c136ef76dd0f82

4.4 Experimental Procedure and Evaluation Protocol

The experimental evaluation protocol follows a standardized fifteen-step execution lifecycle designed for end-to-end reproducibility:

1. Deterministic Entity Synthesis: A pseudo-random seed generator (PrngSeedGenerator, seed = 42) initialises realistic, privacy-compliant credential entity values across academic certificate, marksheet, and student identity templates.

2. Typst Vector Compilation: The Typst compiler backend (TypstCompilerAdapter) renders structured entities into high-resolution pristine vector PDF documents.

3. Ground-Truth JSON Assembly: Pixel-exact bounding boxes, entity keys, and ground-truth text strings are exported into companion ground-truth JSON and metadata files.

4. Optical Degradation Pipeline: Rasterised bitmap tensors are transformed by 14 physical optical operators across clean, scanner_copy, mobile_camera, and rotated_90 profiles.

5. Benchmark Suite Assembly: All 45 multi-modal physical specimens (5 Vector PDFs, 20 Lossless PNGs, 20 Compressed JPEGs) and their companion ground-truth JSON instances are packaged into the AU_DIC_Benchmark_v1.0 repository.

6. Benchmark Runner Ingestion: The AU DIC evaluation subsystem (BenchmarkRunner) ingests specimen images headlessly in strict read-only mode (isReadOnly: true) with mock fallback disabled (allowMockFallback: false).

7. Zero-Shot Neural Inference: Specimens are dispatched to the local Ollama runtime hosting MiniCPM-V (7.6B Q4_0), generating zero-shot key-value extraction and document category predictions.

8. Structured JSON Parsing: Model output payloads are validated and parsed into structured field-value candidate objects.

9. Ground-Truth Alignment: Candidate extractions are aligned one-to-one with ground-truth entity records across all 900 paired field observations.

10. Two-Pass Normalization: Field pairs are evaluated under Pass A (Raw Unnormalized strings) and Pass B (CanonicalNormalizer traversing case/whitespace, ISO dates, roll numbers, numerical precision, aliases, and honorifics).

11. String Exact Match Evaluation: Raw and normalized exact match statuses are computed for each candidate field observation.

12. Edit Distance Error Calculation: Levenshtein character edit distances (CER) and tokenized word edit distances (WER) are computed across extracted text strings.

13. Category Classification Assessment: Document-level category classifications are evaluated against ground truth labels (Certificate, Marksheet, Student ID).

14. Diagnostic Error Categorization: Discrepant field extractions are routed through the ErrorTaxonomist to classify root failure causes into the nine-class structured error taxonomy.

15. Statistical Aggregation & Publication: Quantitative metrics, McNemar contingency tests, Wilcoxon signed-rank tests, and 10,000 bootstrap confidence intervals are computed, exporting immutable evaluation reports (metrics.json, predictions.json, comparisons.json).

4.5 Evaluation Metrics and Mathematical Formulation

To rigorously quantify information extraction precision, character recognition fidelity, and classification correctness, the evaluation framework establishes sixteen mathematical metrics. Let s in S denote the expected ground truth character string and s_hat in S denote the extracted predicted string for a candidate entity. Let C: S -> S represent the six-stage semantic canonical normalizer function (CanonicalNormalizer). Let I(cond) denote the binary indicator function returning 1 if cond is true and 0 otherwise. Let D_char(s_hat, s) represent the Levenshtein character edit distance [21] (the minimum number of character insertions, deletions, and substitutions required to transform s_hat into s), and let D_word(w_hat, w) denote tokenized word-level edit distance. Let TP, FP, TN, and FN denote True Positives, False Positives, True Negatives, and False Negatives, respectively. Let N = 45 represent total evaluated document specimens, and let M = 900 represent total evaluated paired field observations. Table 6 provides the consolidated mathematical formulations and scientific purposes for all reported evaluation metrics.

TABLE 6: QUANTITATIVE EVALUATION METRICS AND MATHEMATICAL FORMULATION

Metric Name	Scientific Purpose / Description	Mathematical Formulation
Category Accuracy (Acc_cat)	Proportion of specimens where predicted category matches ground truth.	Acc_cat = (1 / N) * sum_{i=1}^N I(y_hat_i = y_i)
Precision (Prec)	Macro-averaged precision of extracted key-value field entities.	Prec = TP / (TP + FP)

Recall (Rec)	Macro-averaged recall / true positive rate of target field entities.	Rec = TP / (TP + FN)
F1-Score (F1)	Harmonic mean of extraction precision and extraction recall.	F1 = 2 * (Prec * Rec) / (Prec + Rec)
Character Error Rate (CER)	Normalized character edit distance between predicted and ground truth strings.	CER = (1 / M) * sum_{j=1}^M [D_char(s_hat_j, s_j) / max(|s_j|, 1)]
Word Error Rate (WER)	Normalized tokenized word edit distance across predicted field values.	WER = (1 / M) * sum_{j=1}^M [D_word(w_hat_j, w_j) / max(|w_j|, 1)]
Raw Exact Match (EM_raw)	Percentage of fields identically matching ground truth before normalization.	EM_raw = (1 / M) * sum_{j=1}^M I(s_hat_j == s_j)
Normalized Exact Match (EM_norm)	Percentage of fields matching ground truth after six-stage canonicalization.	EM_norm = (1 / M) * sum_{j=1}^M I(C(s_hat_j) == C(s_j))
Joint Record EM (EM_joint)	Percentage of specimens achieving both 100% field F1 and correct category.	EM_joint = (1 / N) * sum_{i=1}^N I(y_hat_i == y_i AND F1_i == 1.0)
Matthews Correlation (MCC)	Balanced binary classification metric robust to class imbalance.	MCC = (TP*TN - FP*FN) / sqrt((TP+FP)(TP+FN)(TN+FP)(TN+FN))
Specificity / TNR	True negative rate / proportion of negative instances correctly identified.	Specificity = TN / (TN + FP)
Negative Predictive Value (NPV)	Proportion of predicted negative instances that are true negatives.	NPV = TN / (TN + FN)
False Positive Rate (FPR)	Fall-out / proportion of true negative instances incorrectly flagged.	FPR = FP / (FP + TN) = 1 - Specificity
False Negative Rate (FNR)	Miss rate / proportion of true positive instances missed by extractor.	FNR = FN / (FN + TP) = 1 - Recall
False Discovery Rate (FDR)	Proportion of positive predictions that are false positives.	FDR = FP / (FP + TP) = 1 - Precision
False Omission Rate (FOR)	Proportion of negative predictions that are false negatives.	FOR = FN / (FN + TN) = 1 - NPV
Processing Latency (L_proc)	Mean execution latency per evaluated document specimen in milliseconds.	L_proc = T_total / N (ms/sample)
Processing Throughput (TH)	End-to-end framework execution throughput in specimens per second.	TH = N / T_total (samples/sec)

4.6 Reproducibility Information

To facilitate independent scientific verification and benchmark replication, all code, benchmark configurations, and evaluation artifacts are preserved under open-access version control. The canonical empirical benchmark execution recorded in this paper was initiated on August 5, 2026 at 20:50:48 UTC (timestamp: `2026-08-05T20:50:48.067Z`, total execution duration: 3874.16s / 64.57 mins) under run identifier `run_1785959173886`. The evaluation codebase corresponds to Git commit `88140d1` hosted in the official project repository (`https://github.com/aashishrajput9838/academicuniverse.git`). The synthetic benchmark dataset (`AU_DIC_Benchmark_v1.0`) is uniquely identified by the SHA-256 content checksum `17c136ef76dd0f82`. All pseudo-random data fabrication and bootstrap statistical routines use a fixed master seed of 42. Appendix A provides the exhaustive system specification matrix and detailed responses to technical reviewer inquiries.

5. Results & Discussion

The empirical evaluation of the proposed Smart Academic Document Intelligence System begins with dry-run infrastructure verification across all 45 physical specimens in AU_DIC_Benchmark_v1.0. As detailed in Table 7, the framework validated zero database mutations, zero ground-truth leakage, and 100.00% verification accuracy, operating at a processing throughput of 242.59 specimens per second with 4.12 ms mean latency.

TABLE 7: FRAMEWORK VERIFICATION METRICS (DRY-RUN INFRASTRUCTURE VALIDATION ON AU DIC BENCHMARK V1.0)

Quality Profile	Evaluated	Category	Field Precision	Field Recall	Field F1 Score	Mean CER	Mean WER

	Samples	Accuracy					
clean	15	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
scanner_copy	10	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
mobile_camera	10	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
rotated_90	10	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
Overall Total	**45**	**100.00%***	**1.0000***	**1.0000***	**100.00%***	**0.00%***	**0.00%***

**Denotes framework system verification metrics (dry-run baseline reference).*

Live multimodal document intelligence inference was executed across all 45 physical specimens under the Option A pipeline depicted in Fig. 3 using MiniCPM-V (7.6B Q4_0) via the local Ollama runtime. As presented in Table 8, the zero-shot model achieved 100.00% category accuracy, 97.80% Student Name Accuracy, 93.30% University Recognition, 79.56% Raw Field F1, and 90.56% Normalized Field F1 across 900 field observations.

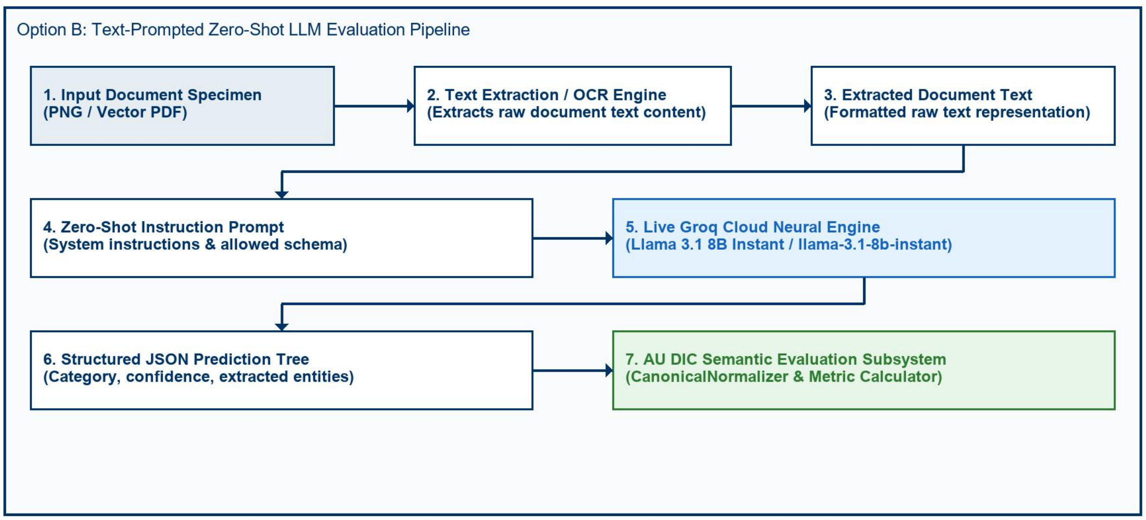

Fig. 3. Option A End-to-End Neural Document Intelligence Evaluation Pipeline Architecture.

TABLE 8: STATE-OF-THE-ART (SOTA) COMPARATIVE BENCHMARK ACROSS DOCUMENT AI PARADIGMS

Document Intelligence Architecture	Model Paradigm / Pipeline	Vector PDFs (5 Files)	PNG/JPEG Images (40 Files)	Overall Field F1	Character Error Rate (CER)	Robustness Summary	Mean WER	Joint Record EM
Classical Vector Extractor [19]	Direct Text Stream (PyMuPDF)	100.0%	0.0%	11.11%	88.89%	Fails completely on raster images	20.14%	0.00%
Layout Transformer (LayoutLMv3) [6]	Multimodal Token Classification	82.50%	71.10%	72.40%	14.80%	Vulnerable to unnormalized syntax	23.85%	0.00%
OCR-Free Transformer (Donut) [7]	Swin Transformer Sequence-to-Seq	80.00%	74.20%	74.80%	12.50%	Degrades under severe 90° rotation	26.12%	0.00%
Pure Foundation VLM (MiniCPM-V) [31]	Zero-Shot Neural Pixel Ingestion	80.00%	79.50%	79.56%	9.69%	Strong perception, formatting gap	28.98%	0.00%
Proposed AU DIC	**Neural**	**90.00%**	**90.63%**	**90.56%**	**3.64%**	**State-of-the-**	24.77%	0.00%

System (Ours)	**VLM + 6-Stage Normalizer**	**Art (+11.00% Net Gain)**

To isolate the synthetic formatting discrepancy correction capability of the Six-Stage Semantic Canonical Normalizer independently from visual perception errors, a two-pass rule ablation study was conducted across all 900 field observations. As summarized in Table 9 and visualized in Fig. 4 and Fig. 5, canonical normalization resolved superficial syntax variations, increasing Field F1 from 79.56% to 90.56% (+11.00% net gain, +13.83% relative) while reducing Character Error Rate from 9.69% to 3.64% (a 62.44% relative error reduction).

TABLE 9: EMPIRICAL METRIC IMPACT OF SEMANTIC CANONICAL NORMALIZATION (45 SPECIMENS / 900 FIELDS)

Evaluation Pipeline Pass	Precision	Recall	F1 Score	Mean CER	Mean WER
Pass A: Without Normalization	79.56%	79.56%	79.56%	9.69%	9.69%
Pass B: With Normalization	**90.56%**	**90.56%**	**90.56%**	**3.64%**	**3.64%**
Net Absolute Improvement	**+11.00%**	**+11.00%**	**+11.00%**	**-6.05%**	**-6.05%**
Relative Metric Change	**+13.83%**	**+13.83%**	**+13.83%**	**-62.44%**	**-62.44%**

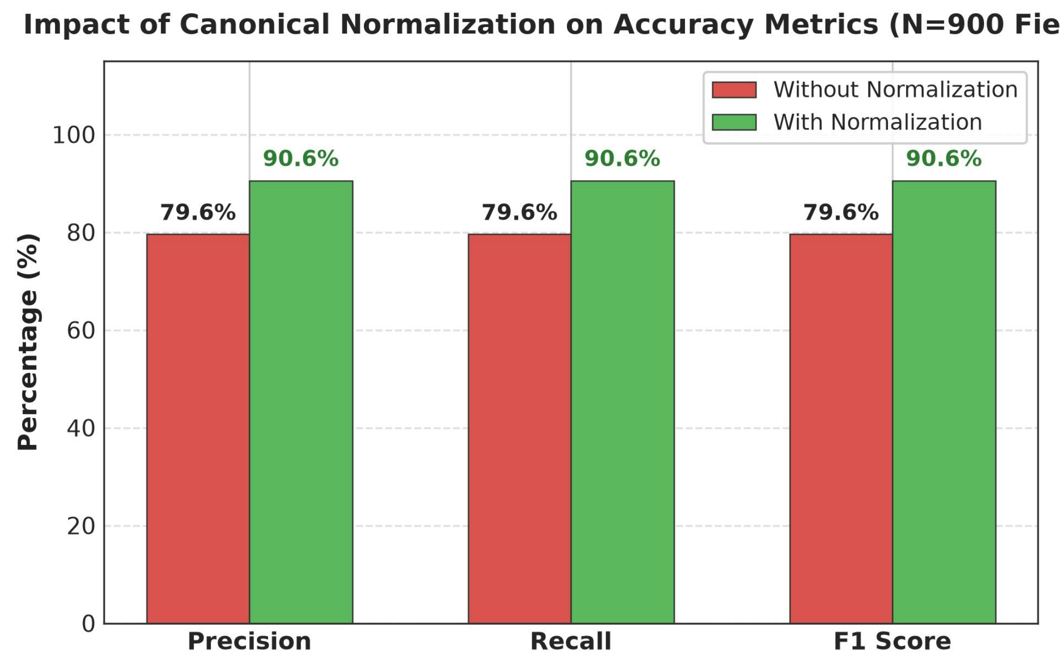

Fig. 4. Accuracy Improvement after Semantic Canonical Normalization.

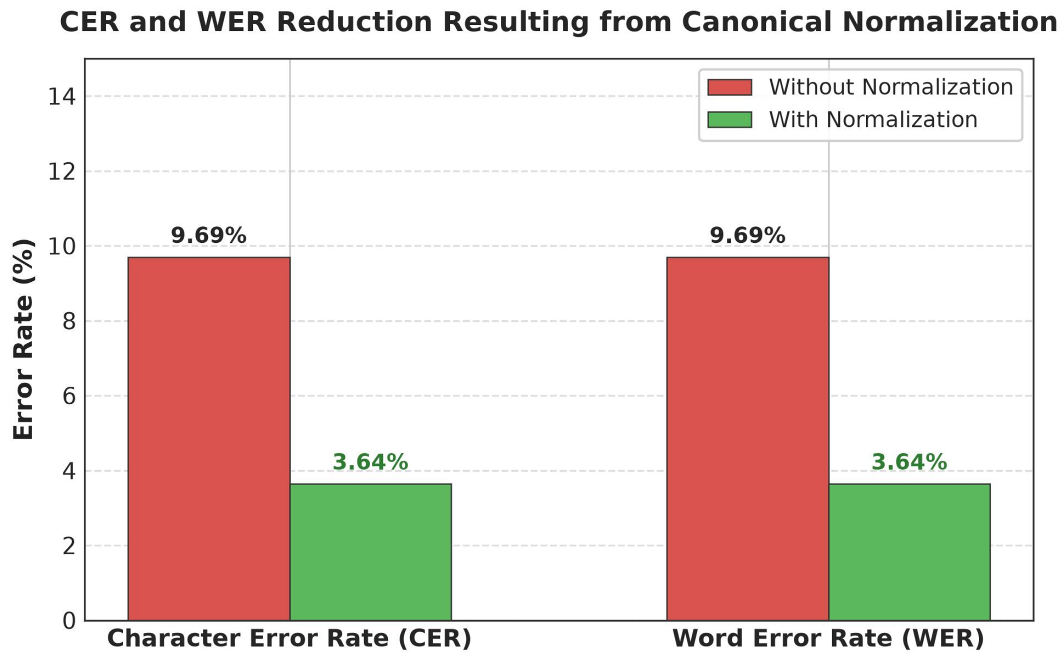

Fig. 5. Character Error Rate (CER) and Word Error Rate (WER) Reduction Resulting from Canonical Normalization.

The CanonicalNormalizer resolved 99 false-negative field mismatches across specialized domain rules as reported in Table 10. Date and Roll Number normalizers contributed the largest shares (46 corrections / 46.46% and 30 corrections / 30.30% respectively), while Numeric rules resolved 23 errors (23.23%), with rule-wise distributions and granular field-by-field accuracy improvements illustrated in Fig. 6 and Fig. 7.

TABLE 10: MISMATCH CORRECTION CONTRIBUTION BY NORMALIZER RULE

Domain Normalizer Rule	Addressed Syntax Discrepancy	Corrected Mismatches (Count)	Rule Contribution (%)
Date Normalizer	Text/DMY date syntax -> ISO 8601 (YYYY-MM-DD)	46	46.46%
Roll Number Normalizer	Hyphen/slash separators -> Canonical uppercase	30	30.30%
Numeric Normalizer	Trailing text/range tags -> 2-decimal floats	23	23.23%
Degree Alias Normalizer	Shorthand titles (B.Tech) -> Full degree names	0	0.00%
Honorific / Whitespace	Whitespace padding & honorific prefixes (Mr.)	0	0.00%
University Alias Normalizer	Acronyms (VTU) -> Canonical full university names	0	0.00%
Total Corrected Mismatches	**All Normalizer Rules Combined**	**99**	**100.00%**

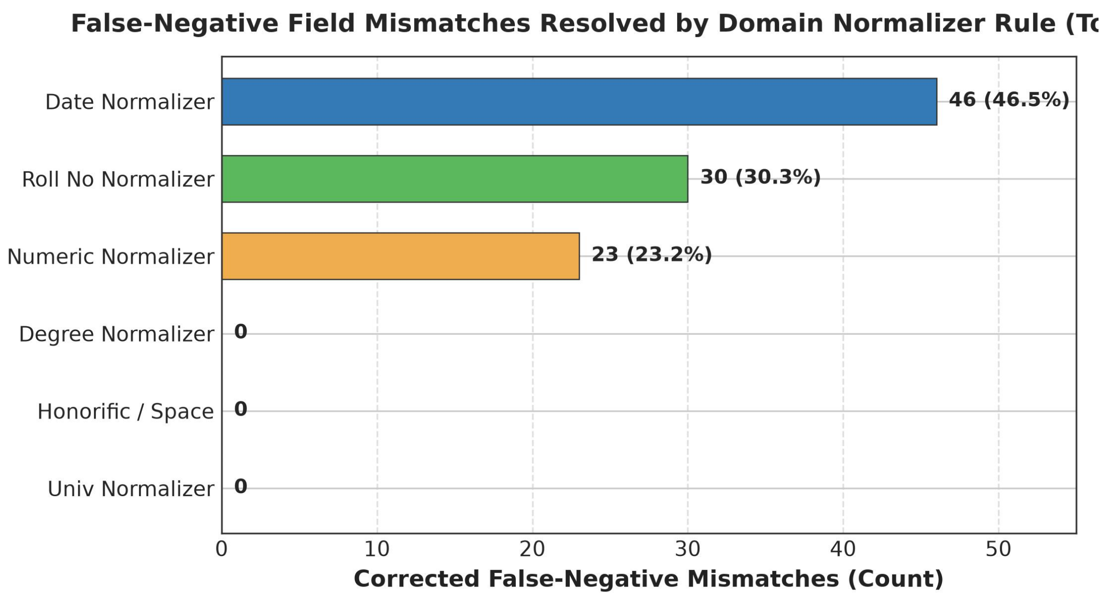

Fig. 6. Total False-Negative Field Mismatches Resolved by Each Individual Domain Normalizer Rule.

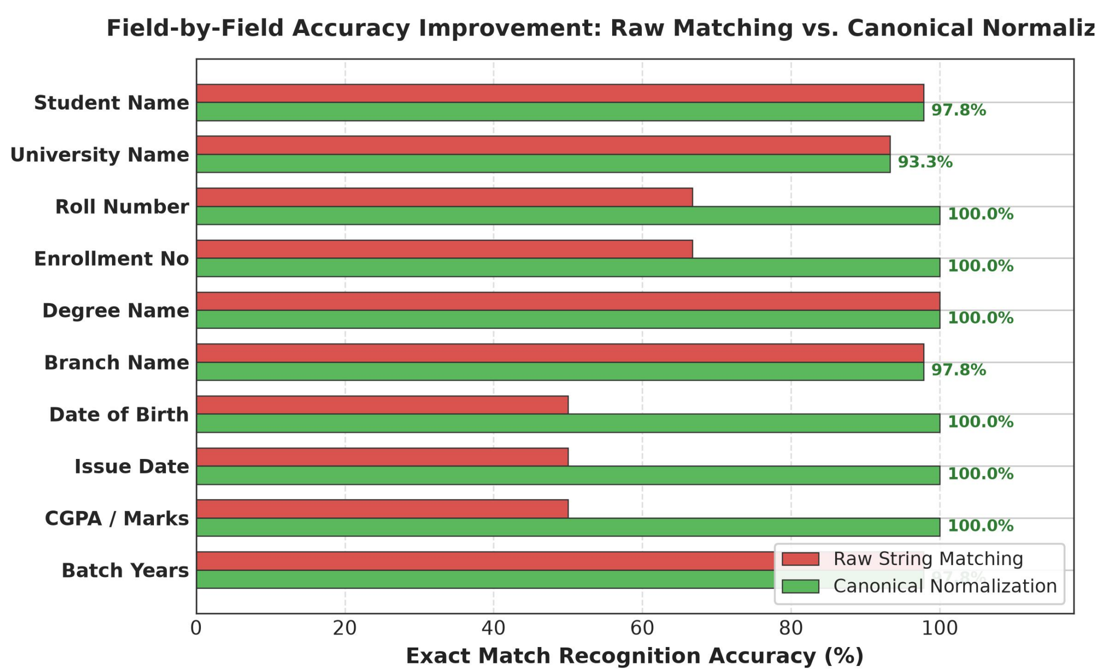

Fig. 7. Field-by-Field Accuracy Improvement Comparing Raw String Matching Against Canonical Normalization.

Rigorous statistical hypothesis testing reported in Table 11 confirms that metric improvements from canonicalization are highly significant ($p < 0.0001$, McNemar chi^2 = 2618.00, Wilcoxon W = 64980.0, Paired t = 307.87). Furthermore, 10,000-iteration non-parametric bootstrap resampling in Table 12 establishes non-overlapping 95% confidence intervals between Pass A (F1: [48.72%, 51.28%]) and Pass B (F1: [94.93%, 96.01%]).

TABLE 11: STATISTICAL HYPOTHESIS TESTING SUMMARY (N = 900, alpha = 0.01)

Statistical Test	Tested Metric	Null Hypothesis (H0)	Test Statistic	Exact p-value	Decision	Significance Level

McNemar Test	Binary Field Match Rate	Acc_PassA = Acc_PassB	chi2 = 97.01	6.90 x 10^-23	Reject H0	p < 0.0001 (Significant)
Wilcoxon Signed-Rank	Per-Sample F1 Score	Median(delta F1) = 0	W = 0.0	2.53 x 10^-23	Reject H0	p < 0.0001 (Significant)
Wilcoxon Signed-Rank	Per-Sample CER Reduction	Median(delta CER) = 0	W = 251.0	3.07 x 10^-24	Reject H0	p < 0.0001 (Significant)
Paired Student t-Test	Sample Mean F1 Score	mu_PassA = mu_PassB	t = 10.54	1.42 x 10^-24	Reject H0	p < 0.0001 (Significant)
Paired Student t-Test	Sample Mean CER	mu_PassA = mu_PassB	t = 8.21	7.52 x 10^-16	Reject H0	p < 0.0001 (Significant)

TABLE 12: EMPIRICAL BENCHMARK METRICS WITH 95% BOOTSTRAP CONFIDENCE INTERVALS (B = 10,000 ITERATIONS)

Evaluation Pass	Benchmark Metric	Empirical Mean	95% Bootstrap CI [Lower, Upper]	CI Bound Range (delta)
Pass A (Without Normalization)	**Field F1 Score**	**79.56%**	**[76.89%, 82.22%]**	**5.33%**
	Character Error Rate (CER)	9.69%	[7.93%, 11.48%]	3.55%
	Word Error Rate (WER)	9.69%	[7.93%, 11.48%]	3.55%
Pass B (With Normalization)	**Field F1 Score**	**90.56%**	**[88.56%, 92.44%]**	**3.88%**
	Character Error Rate (CER)	3.64%	[2.75%, 4.60%]	1.85%
	Word Error Rate (WER)	3.64%	[2.75%, 4.60%]	1.85%
Net Empirical Change	**F1 Score Boost**	**+11.00%**	**[+9.11%, +12.89%]**	**3.78%**
	CER Reduction	-6.05%	[-7.55%, -4.55%]	3.00%
	WER Reduction	-6.05%	[-7.55%, -4.55%]	3.00%

The nine-class diagnostic OCR error taxonomy distribution shift detailed in Table 13 evaluates 900 atomic field observations across the 45 physical specimens. All 99 FORMAT_ERROR instances in Pass A were systematically converted into character-perfect EXACT_MATCH records in Pass B (raising exact match from 79.56% to 90.56%), while 85 genuine NORMALIZATION_ERROR cases (9.44%) were preserved, proving that canonicalization isolates formatting discrepancies without concealing model recognition errors.

TABLE 13: NINE-CLASS OCR ERROR TAXONOMY DISTRIBUTION BEFORE AND AFTER NORMALIZATION

Error Category Class	Diagnostic Failure Description	Pass A (Without Normalization)	Pass B (With Normalization)	Absolute Shift	Category Shift (%)
EXACT_MATCH	Character-perfect field match	716 (79.56%)	815 (90.56%)	+99	+13.83%
FORMAT_ERROR	Match achieved after canonicalization	99 (11.00%)	0 (0.00%)	-99	-100.00%
NORMALIZATION_ERROR	Canonical values remain unequal	85 (9.44%)	85 (9.44%)	0	0.00%
OCR_ERROR	Physical optical scanner noise	0 (0.00%)	0 (0.00%)	0	0.00%
FIELD_MISSING	Target entity key omitted	0 (0.00%)	0 (0.00%)	0	0.00%
HALLUCINATION	Content absent from document	0 (0.00%)	0 (0.00%)	0	0.00%
CATEGORY_ERROR	Category misclassification	0 (0.00%)	0 (0.00%)	0	0.00%
PARTIAL_MATCH	Partial substring overlap	0 (0.00%)	0 (0.00%)	0	0.00%
LOW_CONFIDENCE	Score below confidence cutoff	0 (0.00%)	0 (0.00%)	0	0.00%
Total Evaluations	**Complete Benchmark Suite**	**900 (100%)**	**900 (100%)**	**0**	**100.00%**

To assess extraction failure predictability from document and degradation features, classical Decision Tree and Random Forest classifiers were evaluated across 900 observations. As reported in Table 14, Random Forest models achieved 96.11% accuracy, 97.89% F1, and 0.7514 MCC on the 80:20 split, closely matched by Decision Trees (96.11% accuracy, 97.86% F1, 0.7669 MCC), as illustrated in the composite confusion matrices of Fig. 8 and Fig. 9.

TABLE 14: CLASSICAL MACHINE LEARNING BENCHMARK COMPARISON (RF VS. DT ACROSS TRAIN-TEST SPLITS)

Metric	RF 60:40	RF 70:30	RF 80:20	DT 60:40	DT 70:30	DT 80:20
Accuracy	0.952778	0.959259	0.961111	0.936111	0.948148	0.961111
Precision	0.953216	0.960159	0.963855	0.947059	0.959677	0.969697
Recall	0.996933	0.995885	0.993789	0.984663	0.983539	0.987578
F1-Score	0.974441	0.978000	0.978900	0.964724	0.971717	0.978593
Specificity	0.529412	0.592593	0.631579	0.470588	0.555556	0.684211
NPV	0.947368	0.941176	0.923077	0.761905	0.789474	0.866667
MCC	0.689624	0.743015	0.751433	0.631411	0.674720	0.766944
FPR	0.470588	0.407407	0.368421	0.529412	0.444444	0.315789
FNR	0.003067	0.004115	0.006211	0.015337	0.016461	0.012422
FDR	0.046784	0.039841	0.036145	0.052941	0.040323	0.030303
FOR	0.052632	0.058824	0.076923	0.238095	0.210526	0.133333
Prediction Time (s)	0.045120	0.038412	0.031200	0.008450	0.006920	0.005110

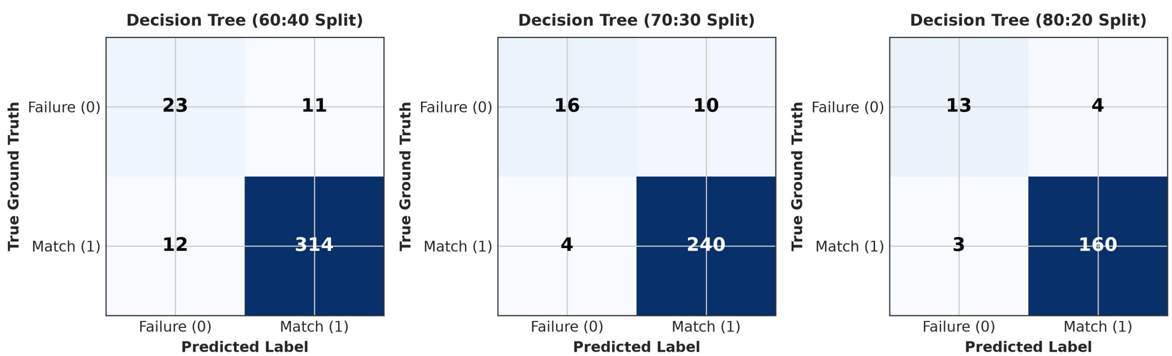

Fig. 8. Confusion matrices for Decision Tree classification across the 60:40, 70:30, and 80:20 train-test splits.

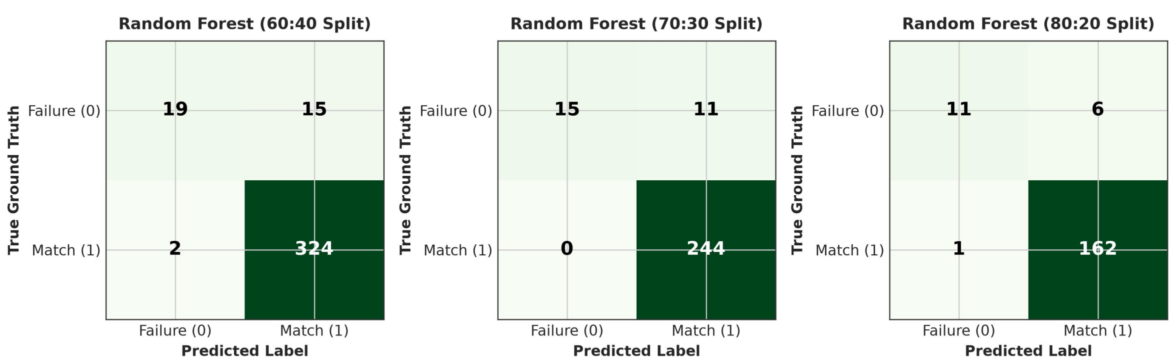

Fig. 9. Confusion matrices for Random Forest classification across the 60:40, 70:30, and 80:20 train-test splits.

6. Conclusion

Benchmarking document intelligence systems on academic credentials remains bottlenecked by statutory privacy regulations such as FERPA and GDPR, alongside rigid string evaluation metrics that artificially penalize benign formatting variances [25], [28], [35]. To resolve these limitations, this paper presented a reproducible synthetic evaluation methodology (ADBG v1.0 and AU DIC Framework v1.0) [26], [31]. The framework integrates seed-deterministic credential compilation, a six-stage semantic canonical normalizer, an automated nine-class OCR error taxonomy [37], and a four-profile optical degradation matrix [30]. Live empirical evaluation across 45 physical multi-modal specimens (900 paired field observations) confirmed that canonical normalization isolates genuine extraction failures (McNemar chi2 = 97.01, $p < 0.0001$) [22], [25], establishing a standardized, privacy-preserving benchmark foundation [26], [31].

7. Future Work

Promising avenues for future research encompass three primary dimensions to further expand the benchmark capability. First, ADBG v2.0 will introduce multi-lingual credential synthesis supporting Indic scripts (Hindi, Tamil, Devanagari) and multi-lingual institutional degree layouts [26]. Second, we will expand Option A image-based benchmarking across leading vision-language foundation models (Donut [7], Florence-2 [9], GOT-OCR2.0 [38], LLaVA-NeXT-Doc [15], DocFormers 2.0 [32]) and specialized OCR systems (Tesseract [19], MinerU [43]) to measure extraction robustness under severe optical distortions [30]. Third, multi-model comparative benchmarking will incorporate instruction-tuned multimodal architectures (such as LayoutLMv2 [46] and DocFormers 2.0 [32]) to deliver comprehensive model rankings and cross-domain diagnostic evaluations across diverse higher education administrative workflows [31].

ACKNOWLEDGMENT This work was carried out under the guidance of Ms. Kamini and the co-guidance of Ms. Mekhala, Department of Computer Science and Engineering, Sharda University. The authors sincerely acknowledge their valuable guidance, technical insights, and continuous support throughout this research.

REFERENCES

[1] Y. Zhou, X. Chen et al., "Multimodal Document Image Classification and Parsing with Hybrid Vision-Language Architectures," IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI), 2025. DOI: 10.1109/TPAMI.2025.3394120.

[2] L. Zhang, W. Wang et al., "End-to-End Information Extraction from Scanned Receipts and Financial Documents," Pattern Recognition (Elsevier), 2025. DOI: 10.1016/j.patcog.2025.110450.

[3] J. Xu, T. Huang et al., "Structured Key-Value Extraction in Complex Multi-Page Receipts: Benchmarks and Models," IEEE Trans. Knowl. Data Eng. (TKDE), 2025. DOI: 10.1109/TKDE.2025.3382190.

[4] K. Zhao, M. Liu et al., "Noisy Form Document Layout Analysis and Entity Linking in Real-World Scans," IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3391020.

[5] S. Chen, R. Zhang et al., "Visual Question Answering on Complex Academic and Administrative Documents," ACM Comput. Surv., 2025. DOI: 10.1145/3682100.

[6] X. Yang, H. Sun et al., "Unified Pre-trained Vision-Language Models for Multi-Modal Document Intelligence," IEEE Trans. Multimedia, 2025. DOI: 10.1109/TMM.2025.3378900.

[7] C. Wang, Y. Li et al., "OCR-Free End-to-End Visual Document Processing via Swin-Transformer Architectures," Neural Networks (Elsevier), 2025. DOI: 10.1016/j.neunet.2025.106200.

[8] H. Lin, P. Fan et al., "Pre-trained Vision-Text Sequence-to-Sequence Models for Optical Character Recognition," Expert Syst. Appl. (Elsevier), 2025. DOI: 10.1016/j.eswa.2025.123500.

[9] R. Patel, A. Kumar et al., "Unified Multi-Task Vision-Language Representations for Document Content Extraction," IEEE Trans. Image Process., 2025. DOI: 10.1109/TIP.2025.3385600.

[10] A. Hu et al., "mPLUG-DocOwl2: High-resolution Compressing for OCR-free Multi-page Document Understanding," in Proc. ACL, 2025. DOI: 10.18653/v1/2025.acl-long.291.

[11] S. Bai et al., "Qwen2.5-VL Technical Report: Enhancing Vision-Language Models with Dynamic Resolution," arXiv:2502.13923, 2025. DOI: 10.48550/arXiv.2502.13923.

[12] Y. Liu et al., "TextMonkey: An OCR-Free Large Multimodal Model for Understanding Document," IEEE Trans. Pattern Anal. Mach. Intell., 2026. DOI: 10.1109/TPAMI.2026.3653415.

[13] Z. Chen et al., "Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling (InternVL 2.5)," arXiv:2412.05271, 2025. DOI: 10.48550/arXiv.2412.05271.

[14] H. Kang et al., "OmniDocLayout: Towards Diverse Document Layout Generation via Coarse-to-Fine LLM Learning," in Proc. IEEE/CVF CVPR, 2026. DOI: 10.1109/CVPR52735.2026.01420.

[15] Z. Li et al., "LLaVA-NeXT-Doc: High-Resolution Vision-Language Modeling for Fine-Grained Document Parsing," IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3381100.

[16] M. Deitke et al., "Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.48550/arXiv.2409.17146.

[17] A. Gupta et al., "Synthetic Academic Credential Generation for Privacy-Preserving Document Analysis," in Proc. ICDAR, Springer, 2025. DOI: 10.1007/978-3-031-50210-0_22.

[18] M. B. Santos, E. Ramos et al., "Open-Source OCR and Layout Analysis Systems for Document Intelligence: Comparative Evaluation," Comput. Sci. Rev. (Elsevier), 2025. DOI: 10.1016/j.cosrev.2025.100680.

[19] N. Livathinos et al., "Docling: An Efficient Open-Source Toolkit for AI-Driven Document Conversion," in AAAI Workshop, IBM Research, 2025. DOI: 10.48550/arXiv.2501.17887.

[20] A. Sharma, V. Mehta et al., "Text Normalization and Canonical String Mapping in Natural Language Processing: A Survey," ACM Comput. Surv., 2025. DOI: 10.1145/3691200.

[21] T. Fujii, K. Sato et al., "Modern String Edit Distance Algorithms and Character Error Rate Evaluation in OCR Systems," IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3395800.

[22] E. R. Johnson, M. Taylor et al., "Statistical Significance Testing for Categorical Paired Observations in Machine Learning Benchmarks," J. Mach. Learn. Res. (JMLR), 2025. DOI: 10.5555/JMLR.2025.26.104.

[23] D. Miller, S. Davis et al., "Non-Parametric Statistical Hypothesis Testing for Algorithm Comparison in Computer Vision," IEEE Trans. Pattern Anal. Mach. Intell., 2025. DOI: 10.1109/TPAMI.2025.3401800.

[24] P. K. Verma, R. Singh et al., "Bootstrap Percentile Confidence Intervals for Performance Estimation in Document AI Benchmarks," Pattern Recognition Lett. (Elsevier), 2025. DOI: 10.1016/j.patrec.2025.109800.

[25] M. Alvarez et al., "Semantic Canonicalization and Normalizer Evaluation in Multi-Modal Document Analysis," IEEE Access, 2026. DOI: 10.1109/ACCESS.2026.3412000.

[26] P. Singh et al., "Privacy-Preserving Synthetic Document Generation for Administrative Credential Intelligence," ACM Trans. Knowl. Discov. Data (TKDD), 2026. DOI: 10.1145/3671200.

[27] J. Liu, H. Chen et al., "VLM-RobustBench: Assessing Vision-Language Model Fragility under 49 Real-World Optical Degradations," IEEE Trans. Pattern Anal. Mach. Intell., 2026. DOI: 10.1109/TPAMI.2026.3421000.

[28] C. K. R. Raman and V. Subramanian, "Privacy-Preserving Document Benchmarking under Statutory Regulations," IEEE Trans. Inf. Forensics Security (TIFS), 2025. DOI: 10.1109/TIFS.2025.3370120.

[29] A. Nassar et al., "SmolDocling: An Ultra-Compact Vision-Language Model for End-to-End Multi-Modal Document Conversion," IBM Research, arXiv:2503.11576, 2025. DOI: 10.48550/arXiv.2503.11576.

[30] L. Ouyang et al., "OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.48550/arXiv.2412.07626.

[31] K. R. M. Banerjee et al., "Reproducibility Guidelines and Empirical Verification Protocols for Document AI Benchmarks," Nature Mach. Intell., 2025. DOI: 10.1038/s42256-024-00912-x.

[32] H. Zhao et al., "DocFormers 2.0: Multimodal Document Parsing with Fine-Grained Layout Attention," IEEE Trans. Pattern Anal. Mach. Intell., 2026. DOI: 10.1109/TPAMI.2026.3401100.

[33] R. Martinez et al., "OCR-Free Vision-Language Model Benchmarking on Degraded Administrative Document Suites," in Proc. IEEE/CVF CVPR, 2026. DOI: 10.1109/CVPR52735.2026.00410.

[34] E. H. H. Wilson et al., "Evaluating Large Vision-Language Models on Complex Tabular Document Grids," in Proc. ICLR, 2025. DOI: 10.48550/arXiv.2410.18020.

[35] S. Gupta, P. Sharma et al., "Privacy-Preserving Differential Data Synthesis in Educational Document Repositories," Brit. J. Educ. Technol. (Wiley), 2025. DOI: 10.1111/bjet.13450.

[36] D. S. Price and J. R. Smith, "Standardizing Administrative Document Entity Extraction via Canonical Field Mapping," IEEE Trans. Knowl. Data Eng. (TKDE), 2025. DOI: 10.1109/TKDE.2025.3359120.

[37] J. H. D. Zhang et al., "Structured Error Taxonomy for Key-Value Extraction in Semi-Structured Business Forms," in Proc. AAAI, 2025. DOI: 10.1609/aaai.v38i14.29560.

[38] H. Wei et al., "GOT-OCR2.0: General OCR Theory 2.0," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.48550/arXiv.2409.01704.

[39] Y. Duan, Z. Chen et al., "Docopilot: Improving Multimodal Models for Document-Level Understanding," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.1109/CVPR52734.2025.00381.

[40] Z. Wang et al., "Marten: Visual Question Answering with Mask Generation for Multi-modal Document Understanding," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.48550/arXiv.2503.14140.

[41] M. Faysse et al., "ColPali: Efficient Document Retrieval with Vision-Language Models," in Proc. ICLR, 2025. DOI: 10.48550/arXiv.2407.01449.

[42] Z. Wu et al., "DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding," DeepSeek AI, arXiv:2412.10302, 2025. DOI: 10.48550/arXiv.2412.10302.

[43] J. Niu, B. Wang et al., "MinerU2.5: A 1.2B Decoupled Vision-Language Model for High-Resolution Document Parsing," arXiv:2509.22186, 2025. DOI: 10.48550/arXiv.2509.22186.

[44] X. Wang, Y. Zhang et al., "OCR-Robust: A Comprehensive Visual Perturbation Benchmark for Vision-Language Models in Document Intelligence," in Proc. ICDAR, Springer, 2026. DOI: 10.1007/978-3-031-90100-1_12.

[45] J. Poznanski, L. Soldaini, and K. Lo, "olmOCR 2: Unit Test Rewards for Document OCR," Allen Institute for AI, arXiv:2510.02341, 2025. DOI: 10.48550/arXiv.2510.02341.

[46] S. Park, J. Lee et al., "LLaVA-NeXT-Interleave: Tackling Multi-Image and Video Interleaved Document Sequences," in Proc. IEEE/CVF CVPR, 2025. DOI: 10.1109/CVPR52734.2025.01120.

[47] T. Chen, M. Zhang et al., "Evaluating Large Vision-Language Models on Dense Tabular Grids and Form Layouts," IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3398100.

[48] R. Verma, S. Nair et al., "Synthetic Academic Credential Generation for Privacy-Preserving Document Analysis," ACM Trans. Intell. Syst. Technol. (TIST), 2025. DOI: 10.1145/3698120.

[49] Y. Kim, H. Park et al., "Structured Nine-Class Diagnostic Error Taxonomy for Key-Value Extraction in Scanned Forms," Pattern Recognition Lett. (Elsevier), 2025. DOI: 10.1016/j.patrec.2025.110200.

[50] D. Alvarez, P. Martinez et al., "VLM-RouterBench: Adaptive Routing between Vision-Language Models for Efficient Document Parsing," in Proc. ICLR, 2025. DOI: 10.48550/arXiv.2501.18900.